

PRACTICE EXERCISES



Module 4: Constraints, Database Design & Normalization

Module 4 - 8 Hours

Constraints, Database Design & Normalization

Instructions

Work against `icicibank` with the full dataset and the `branch_customer_flat` table loaded.

Level	Meaning	Expected time
Level 1	Recall and understanding	30 min
Level 2	Apply. Design, build and normalize.	65 min
Level 3	Judgement. Defend a design decision.	35 min

Rules:

- Design answers must be written as `CREATE TABLE` statements with every constraint named.
- Where a question asks "why", one sentence of reasoning is worth more than three of description.
- Prefix any table you create with `ex4_` so it is easy to clean up.

How much to attempt

The questions marked **(Core)** are the required set. They fit the time above, and together they cover every objective of the module. Everything unmarked is **stretch work**: attempt it if you finish early, or take it home.

You are not behind if you have completed only the core set.

Level	Core set	Stretch
Level 1	Q1, Q2, Q3, Q4, Q7, Q9, Q10	Q5, Q6, Q8
Level 2	Q11, Q14, Q15, Q16, Q17, Q19, Q21	Q12, Q13, Q18, Q20
Level 3	Q23, Q25	Q22, Q24, Q26

Level 1 - Understanding

Q1. (Core) For the `loan` table, list one example of each: a super key, a candidate key, the primary key, and a foreign key. State why your super key is not a candidate key.

Q2. (Core) ICICI Bank proposes using `email` as the primary key of `customer`. Give three separate reasons to refuse, and say what you would do instead so the business rule that emails are unique is still enforced.

Q3. (Core) Explain the difference between `RESTRICT` and `CASCADE`, then decide the correct rule for each of these and give a one-line reason.

1. `account_txn` referencing `account`
2. `card_reward_enrolment` referencing `credit_card`
3. `loan` referencing `customer`
4. `employee.manager_id` referencing `employee`
5. `loan_payment` referencing `loan`

Q4. (Core) Define the insert, update and delete anomaly. For each, give the specific example you observed in `branch_customer_flat` during the lab.

Q5. Read the crow's foot notation and write out in words what it says, including optionality on both sides.

```
BRANCH ||-----o{ EMPLOYEE
LOAN   ||-----|{ LOAN_PAYMENT
```

Q6. Explain why a many-to-many relationship cannot be stored with a foreign key, and describe what a junction table's primary key should be and why.

Q7. (Core) State the rule for each normal form and the anomaly it removes.

Form	Rule	Anomaly removed
1NF	?	?
2NF	?	?
3NF	?	?
BCNF	?	?

Q8. Why can a table with a single-column primary key never violate 2NF? What does that imply about a schema that uses surrogate keys throughout?

Q9. (Core) For each, name the normal form violated and the fix.

1. `customer(customer_id, name, phone_1, phone_2, phone_3)`
2. `order_line(order_id, product_id, quantity, product_name, product_price)`
3. `employee(employee_id, name, branch_id, branch_name, branch_city)`
4. `account(account_id, customer_id, customer_name, balance)`

Q10. (Core) `account.balance` is derivable from `account_txn` and is stored anyway. Name the normalization rule this breaks, give the business reason it is stored, and state the one thing that must exist alongside it.

Level 2 - Design and normalize

Analyse the flat table

Q11. (Core) Write a query proving that `branch_ifsc` is functionally dependent on `branch_code` in `branch_customer_flat`, and another proving that `branch_name` is not consistently so. Explain what the second result means.

Q12. Write a query that reports, for each branch code, how many times the branch's details are redundantly repeated, and how many bytes of `branch_name` are stored in total against how many would be stored if it appeared once.

Q13. Show that `cust_phone` is a candidate key of `branch_customer_flat` in the current data. Then explain, in one sentence, why you would still not use it as the primary key.

Normalize

Q14. (Core) Decompose `branch_customer_flat` into 1NF. Produce `ex4_customer_account` from the repeating group and `ex4_customer_product` from the multi-valued column. State the row count of each.

Q15. (Core) Decompose to 3NF. Produce `ex4_branch` and `ex4_customer` with the correct primary and foreign keys declared. Handle the four spellings of BR03 and state, in one sentence, the rule you applied to choose the surviving name.

Q16. (Core) Prove all three anomalies are gone from your decomposed design. Write one statement per anomaly, each demonstrating that the operation now succeeds or now fails safely, and show the output.

Design from requirements

Q17. (Core) ICICI Bank is launching a locker service. Design the schema from this requirement. Name every constraint and choose every cascading rule deliberately.

A branch has a number of lockers, each with a size and an annual rent. A locker may be allotted to one customer at a time, under an agreement with a start date, an end date and a status. A customer may hold lockers at more than one branch. Every visit to a locker is logged with the date, time, and which employee supervised it. A locker that has never been allotted must still be recorded so the branch knows its inventory.

Q18. For your locker design, answer in a table: which entity is weak and why; which relationship is many-to-many and how you resolved it; which foreign key is nullable and why; and which two business rules you expressed as `CHECK` constraints.

Q19. (Core) Add a `CHECK` constraint to your locker agreement table that prevents an end date on or before the start date. Then write the `INSERT` that proves it works.

Denormalize

Q20. Design a `ex4_branch_daily_summary` table that stores, per branch per day, the transaction count, total credits, total debits and closing deposit total. State the primary key, and write the `INSERT ... SELECT` that populates it from the transactional tables for January 2026.

Q21. (Core) Write the reconciliation query that proves your summary table agrees with the transactional tables, and run it. State what a non-empty result would mean.

Level 3 - Judgement

Q22. A vendor's package uses `ON DELETE CASCADE` on every foreign key in the schema, including `account_txn` referencing `account` and `account` referencing `customer`. The vendor says this "keeps the database clean". Write the response you would send to the vendor and to your own head of data governance. State the specific scenario you are worried about and what you require instead.

Q23. (Core) An architect proposes storing a customer's up to three phone numbers as `phone_1`, `phone_2`, `phone_3` rather than a separate `customer_phone` table, arguing that "nobody has more than three, and the join costs us performance on the customer screen". Give two arguments against and one situation in which their position would actually be reasonable.

Q24. ICICI Bank's reporting warehouse stores `branch_name` on every row of a 400-million-row fact table, rather than joining to a branch dimension. Explain why this is acceptable there but would not be acceptable in `customer`, and name the control that makes it safe.

Q25. (Core) During the flat-table cleanup you discovered four spellings of one branch name. Your team lead suggests picking the most frequent spelling automatically for all such cases and moving on, because there are 8 branches and a deadline. Write your response. If you would accept the shortcut anywhere, say exactly where and what you would require alongside it.

Q26. A colleague argues that normalization is "an academic exercise" because modern hardware makes joins cheap and storage is free. Take the argument seriously and respond to it on its own terms. What is the real cost that normalization prevents, and is it a storage cost?

Q27. (Written, 10 to 15 lines.) Reply to ticket ICICI-4577 for the Head of Data Governance. Explain what was wrong with the customer master table in terms a non-technical governance lead will follow, what you have replaced it with, how each of the two audit findings is now impossible rather than merely unlikely, and what the branches will have to do differently. Do not use the words "normalization", "1NF", "2NF", "3NF" or "transitive dependency".